



ORACLE

MySQL Implementation Essentials Bootcamp

MySQL HeatWave

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Program Agenda

- Introduction
- Oracle MySQL Enterprise
- Oracle Cloud Infrastructure overview
- Installation and Architecture
- Database Design
- Security
- Backup
- Replication
- High Availability
- MySQL Enterprise Monitor
- **HeatWave**

MySQL HeatWave

“

*The world's two most popular databases are the Oracle Autonomous Database and Oracle MySQL," said Oracle Chairman and CTO, Larry Ellison. "The Oracle Database once again delivered solid revenue growth in FY21. **And while our Oracle Database business as measured by revenue currently dwarfs our MySQL database business—that is about to change because the latest version of Oracle MySQL has been upgraded to include a revolutionary new ultra-high-performance parallel processing query engine called HeatWave.***

*Independent analysts have tested and confirmed that Oracle MySQL with HeatWave runs 10 to 100 times faster than Amazon's version of MySQL called Aurora. This technological breakthrough is causing several of Amazon's customers to **start moving their Aurora workloads to Oracle MySQL.** And industry analysts are telling us they are seeing a 10x increase in Oracle Cloud Infrastructure customer inquiries.*

*Both the Oracle Autonomous Database and Oracle MySQL with HeatWave technology have **captured the technology high-ground in the cloud database business—and that bodes well for the future of the Oracle Cloud.***

Larry Ellison
Chairman, CTO, Oracle

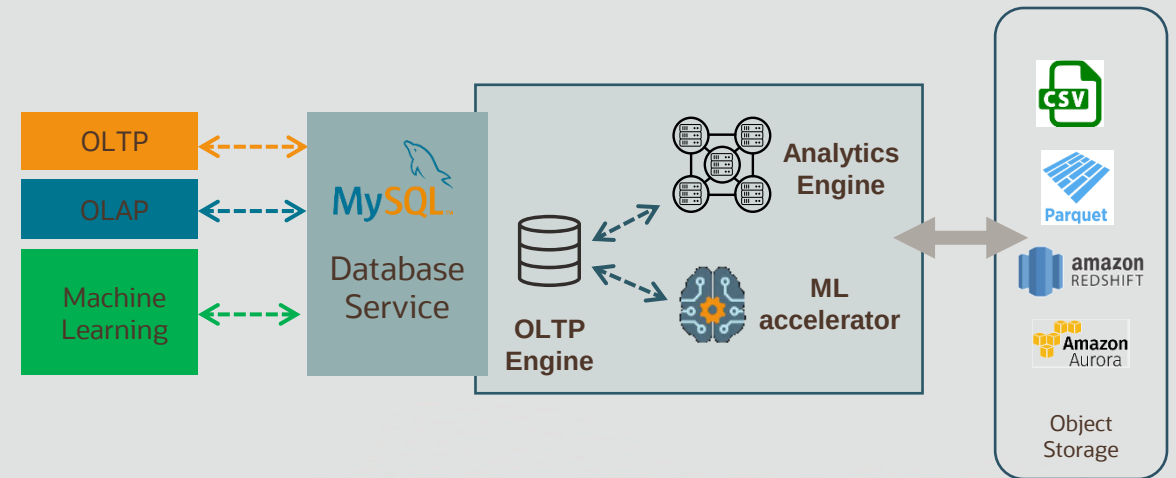
Larry Ellison, Chairman, CTO, Oracle
June 15, 2021

MySQL HeatWave - Overview

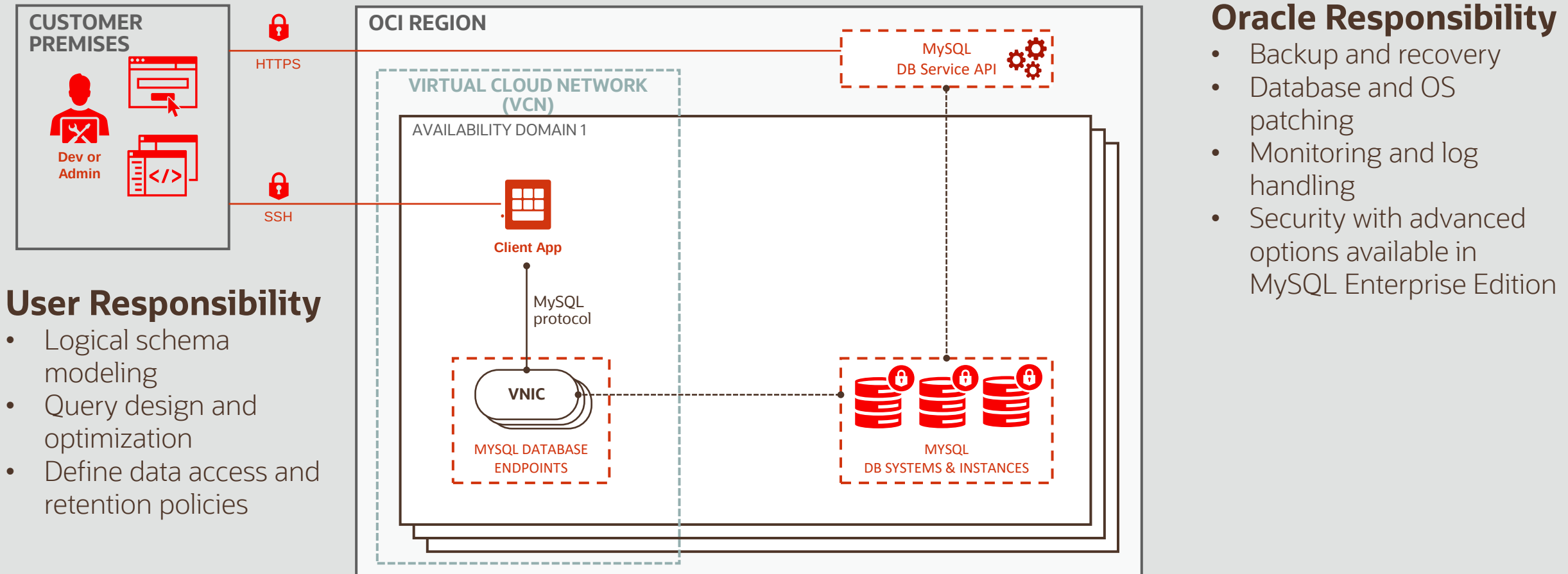
- **Fully-managed** MySQL Service & very easy to use
 - Deploy, Backup&Restore, High Availability (RPO=0), Resize, Read Replicas...
- Powerful union of **Oracle Cloud Infrastructure (Gen2)** and **MySQL Enterprise Edition 8**
- **Always up to date** with the latest security fixes

- **Use cases**

- OLTP only
 - Analytics
 - Query accelerator (OLTP+OLAP)
 - Machine Learning
 - LakeHouse (direct access to Object Store)
- No need of ELT/ETL
 - Great price/performance ratio
 - <https://blogs.oracle.com/mysql/post/mysql-cloud-services-cost-comparison-who-provides-the-best-value>
 - The **only** MySQL Database Service with **Oracle Premier Support for MySQL**



Managed MySQL: Focus on Your Business



MySQL HeatWave Database Service Shapes

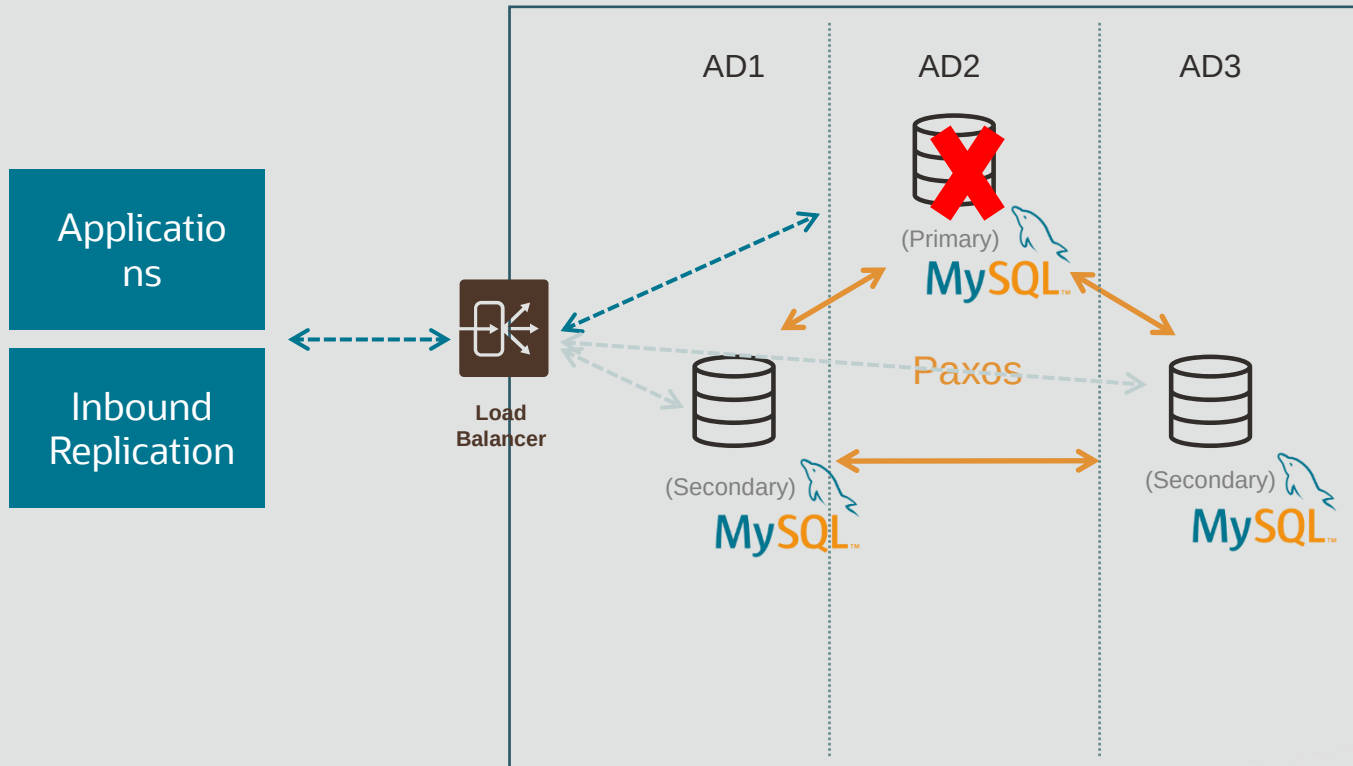
\$43/month + 50GB disk

OCPUs	RAM (GB)	AMD E3 (VM.Standard.E3)	AMD E4 (VM.Standard.E4)	Intel X9 (VM.Standard3)	Intel X9 (VM.Optimized3)	Intel X (VM.Standard2)
1	8	✓	✓	✓	✓	
1	16	✓	✓	✓	✓	✓
2	32	✓	✓	✓	✓	✓
4	64	✓	✓	✓	✓	✓
8	128	✓	✓	✓	✓	✓
16	256	✓	✓	✓	✓	✓
24	384	✓	✓	✓		
32	512	✓	✓	✓		
48	768	✓	✓			
64	1024	✓	✓			

\$3,485/month + disk

- Starting with 50 GB (Storage Size)
- Up to 128.000 GB
- Backup (up to Storage Size is for free)

High Availability



Single Click High Availability

Automatic Failover

Based on MySQL Group Replication

Deployed across fault and availability domains

Seamlessly integrated with replication into MySQL Service

No data loss in case of failure (RPO=0)

Online and fast fail-over (RTO < 2min)

Standalone

Single-instance MySQL DB System

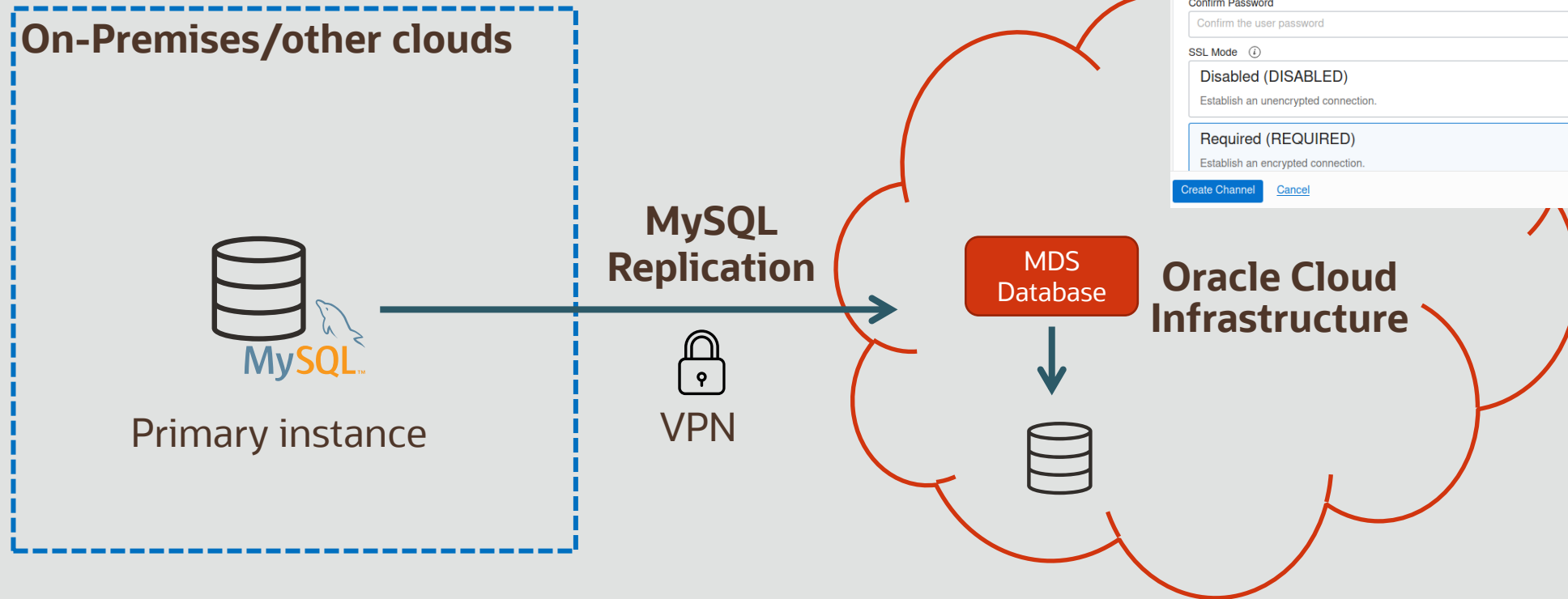
High Availability

Run 3-instance MySQL DB System providing automatic failover and zero data loss

HeatWave

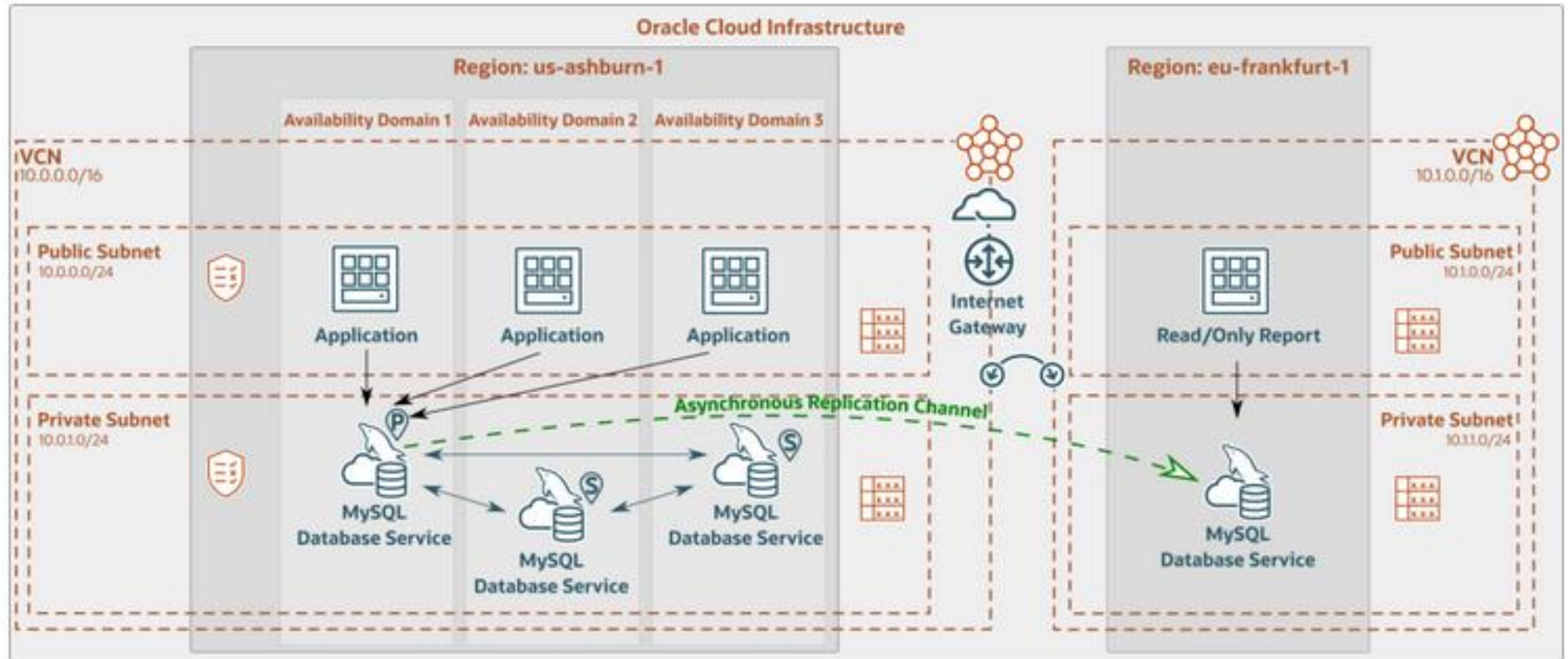
DB System that allows you to enable HeatWave for accelerated query processing, suitable for running both OLTP and OLAP workloads

Inbound Replication



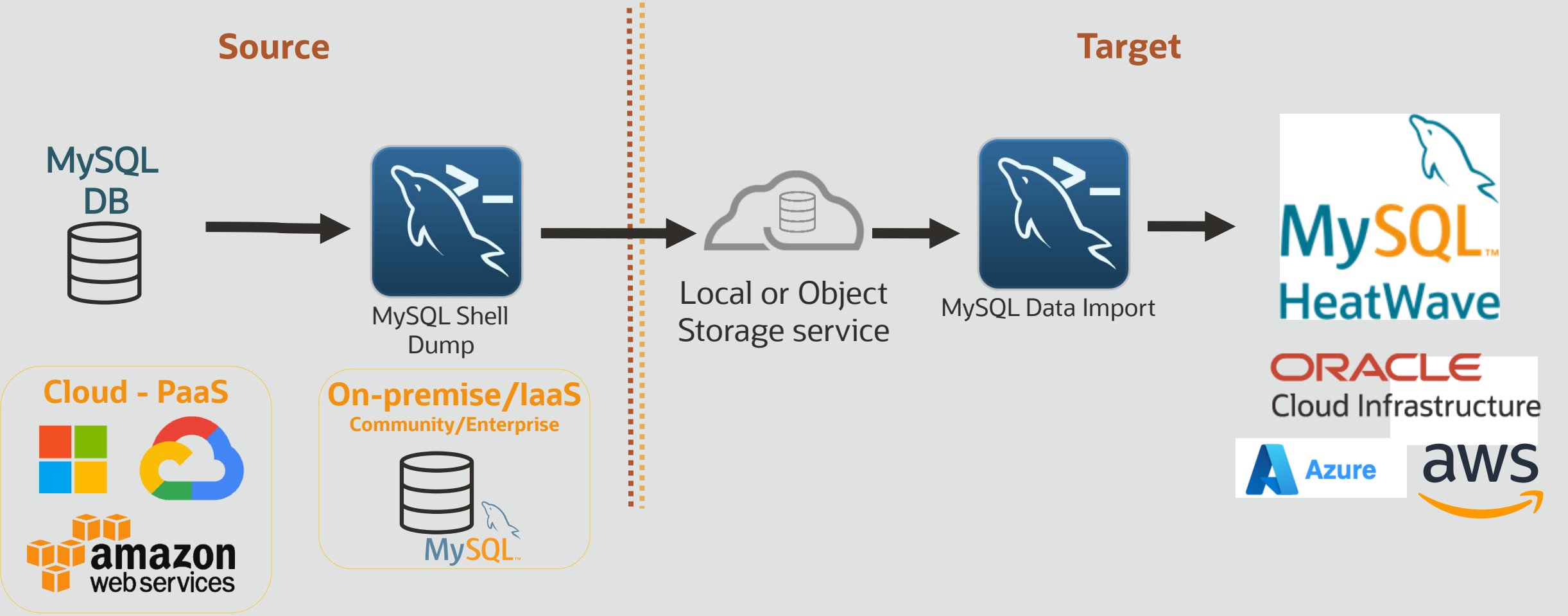
- Multi-source replication not supported
- Many filters pre-configured (also replication from non-GTID sources)

Outbound Replication

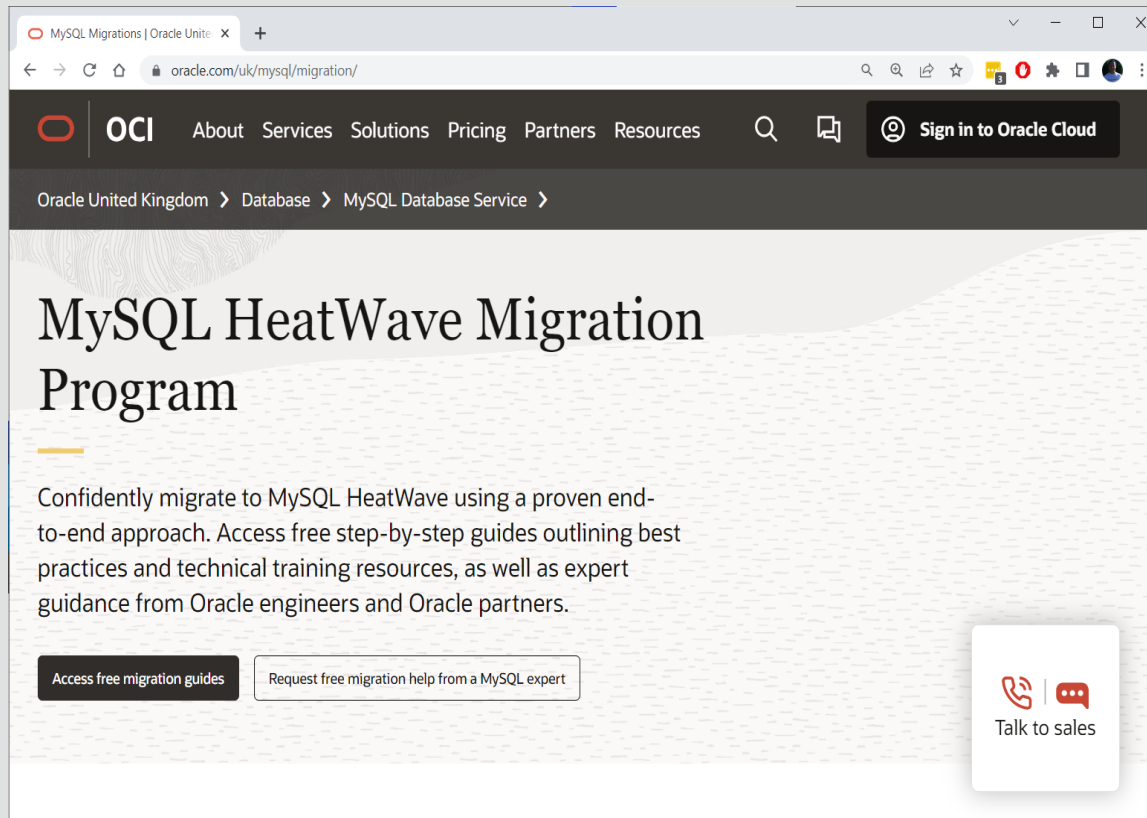


MySQL Migration - tools and a team of experts available

MySQL Shell Dump & Load



Migrating to MySQL HeatWave



A screenshot of the Oracle MySQL Migration Program page. The browser address bar shows 'oracle.com/uk/mysql/migration/'. The page features a dark navigation bar with the OCI logo and links for About, Services, Solutions, Pricing, Partners, and Resources. Below this is a breadcrumb trail: 'Oracle United Kingdom > Database > MySQL Database Service >'. The main heading is 'MySQL HeatWave Migration Program'. The text describes a confident migration process using a proven end-to-end approach, offering free step-by-step guides, technical training resources, and expert guidance from Oracle engineers and partners. At the bottom, there are two buttons: 'Access free migration guides' and 'Request free migration help from a MySQL expert'. A 'Talk to sales' chat bubble is visible in the bottom right corner.

MySQL Migrations | Oracle Unite x +

oracle.com/uk/mysql/migration/

OCI About Services Solutions Pricing Partners Resources Sign in to Oracle Cloud

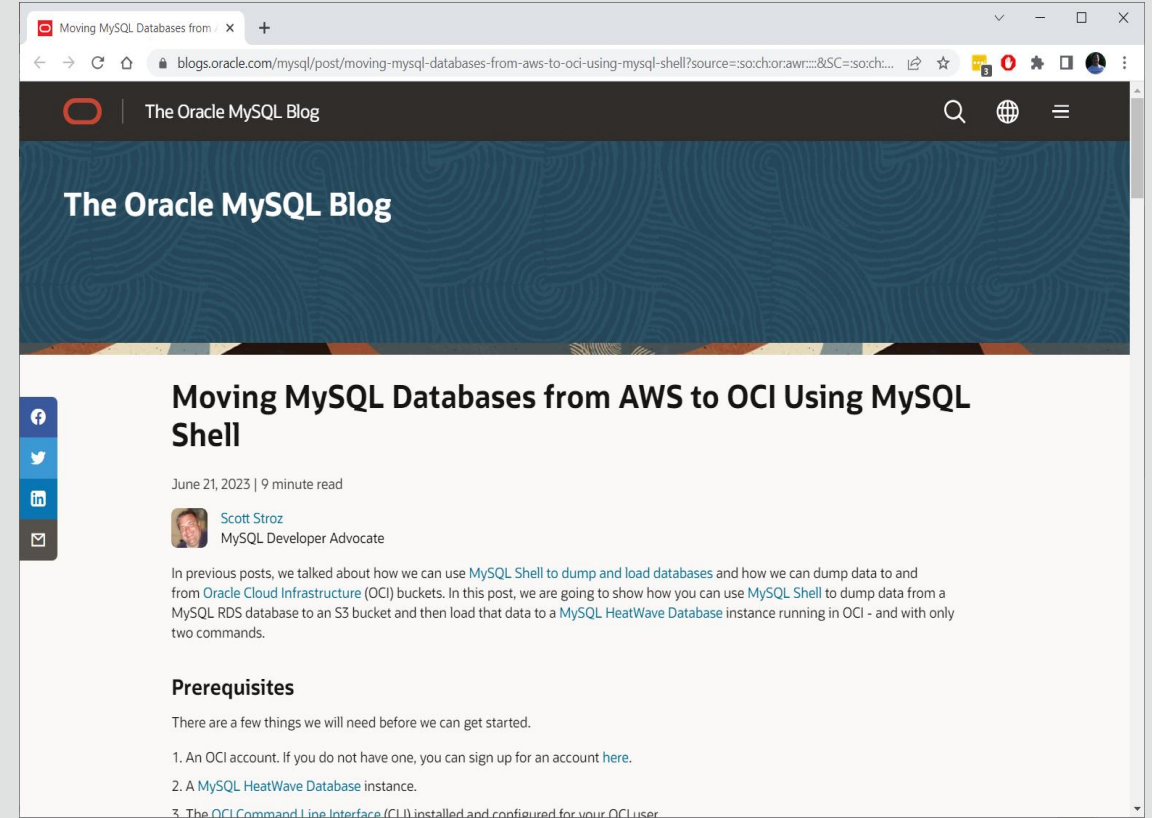
Oracle United Kingdom > Database > MySQL Database Service >

MySQL HeatWave Migration Program

Confidently migrate to MySQL HeatWave using a proven end-to-end approach. Access free step-by-step guides outlining best practices and technical training resources, as well as expert guidance from Oracle engineers and Oracle partners.

Access free migration guides Request free migration help from a MySQL expert

Talk to sales



A screenshot of a blog post on 'The Oracle MySQL Blog'. The browser address bar shows 'blogs.oracle.com/mysql/post/moving-mysql-databases-from-aws-to-oci-using-mysql-shell?source=so:ch:or:aw:;:&SC=:so:ch:'. The page has a dark navigation bar with the OCI logo and 'The Oracle MySQL Blog' title. The main heading is 'Moving MySQL Databases from AWS to OCI Using MySQL Shell'. The post is dated 'June 21, 2023 | 9 minute read' and is by 'Scott Stroz, MySQL Developer Advocate'. The text describes how to use MySQL Shell to dump and load databases from Oracle Cloud Infrastructure (OCI) buckets to a MySQL HeatWave Database instance. It includes a 'Prerequisites' section with three items: 1. An OCI account, 2. A MySQL HeatWave Database instance, and 3. The OCI Command Line Interface (CLI) installed and configured for your OCI user.

Moving MySQL Databases from x +

blogs.oracle.com/mysql/post/moving-mysql-databases-from-aws-to-oci-using-mysql-shell?source=so:ch:or:aw:;:&SC=:so:ch:...

The Oracle MySQL Blog

The Oracle MySQL Blog

Moving MySQL Databases from AWS to OCI Using MySQL Shell

June 21, 2023 | 9 minute read

Scott Stroz
MySQL Developer Advocate

In previous posts, we talked about how we can use [MySQL Shell to dump and load databases](#) and how we can dump data to and from [Oracle Cloud Infrastructure \(OCI\) buckets](#). In this post, we are going to show how you can use [MySQL Shell](#) to dump data from a MySQL RDS database to an S3 bucket and then load that data to a [MySQL HeatWave Database](#) instance running in OCI - and with only two commands.

Prerequisites

There are a few things we will need before we can get started.

1. An OCI account. If you do not have one, you can sign up for an account [here](#).
2. A [MySQL HeatWave Database](#) instance.
3. The [OCI Command Line Interface \(CLI\)](#) installed and configured for your OCI user.





Hands-On Labs



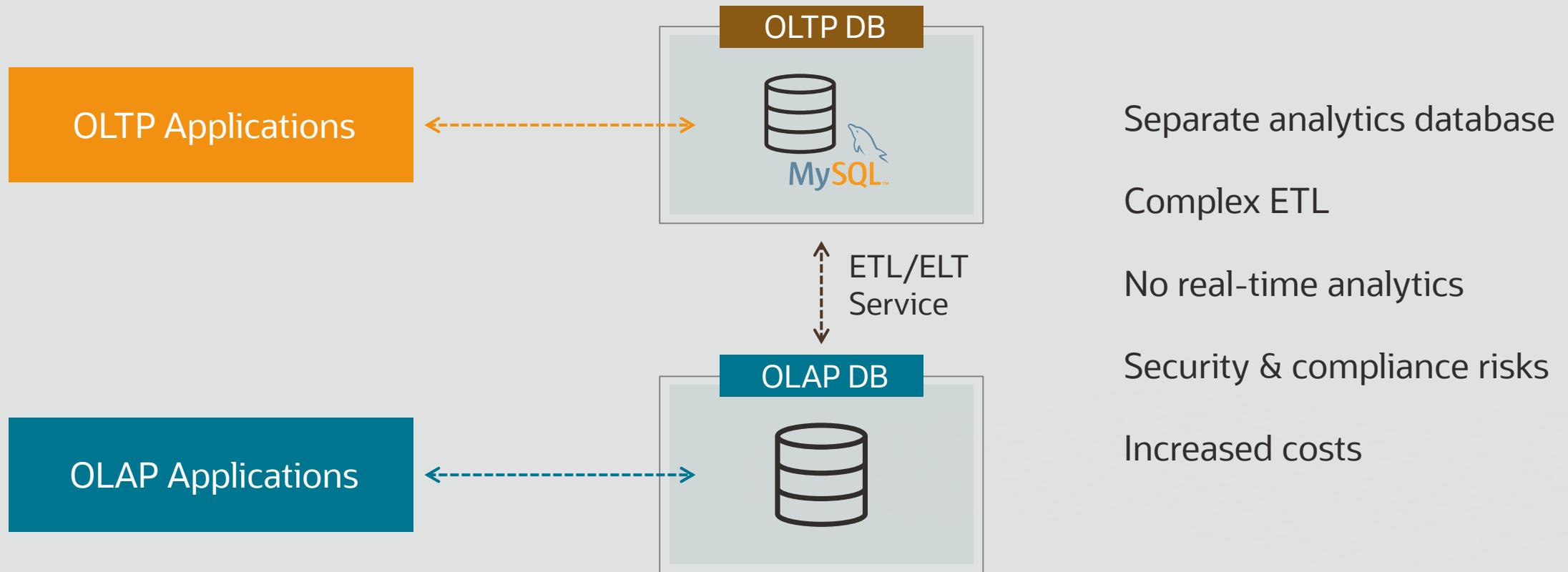
9a. DEMO: MySQL HeatWave Database Service

HeatWave

A short reminder: What's MySQL?

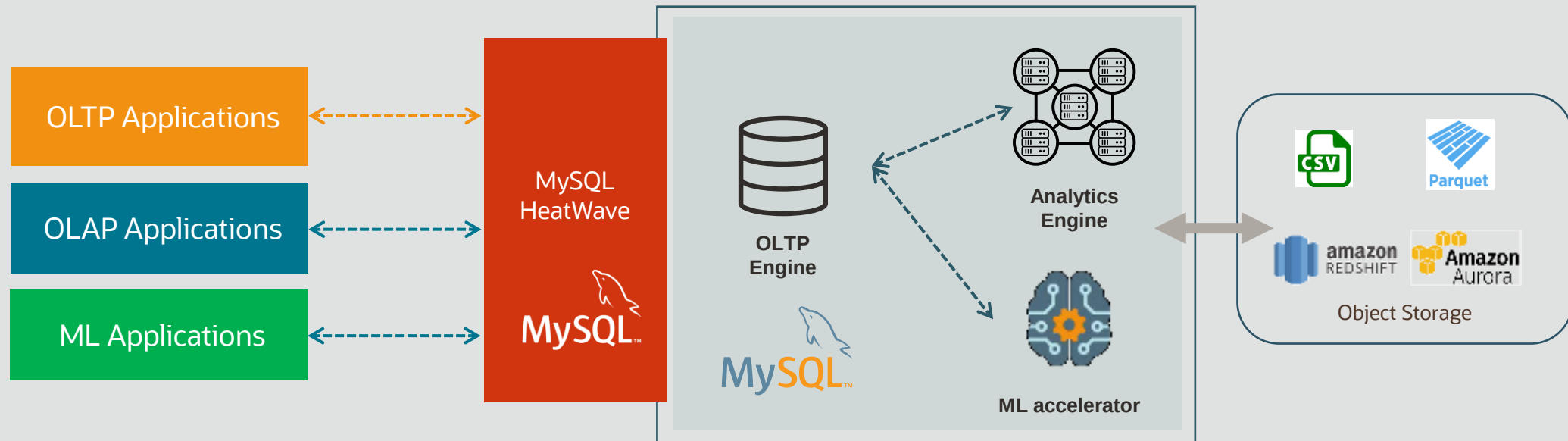
- Relational Database...
- Transactional, ACID, MVCC – ARIES
- OLTP: low latency, high throughput
- InnoDB supports encryption and compression and InnoDB Cluster
- A typical MySQL database: Up to some TBs
- Theoretical limits – maximum tablespace size is 64TB (256TB)

Problem: MySQL users need separate systems for OLTP and OLAP



HeatWave for real-Time Analytics

- Single MySQL database for OLTP, OLAP/Analytics and Machine Learning applications
- All **existing applications** will work **without any changes** – incl. Tableau, Qlik, Looker
- Machine learning based **automation**
- **Extreme performance**: Accelerates MySQL by orders of magnitude up to
- Direct access to Object Storage for **Lakehouse**
- Starting with **281.45\$/month** (for ~50Gb data incl. Backup)



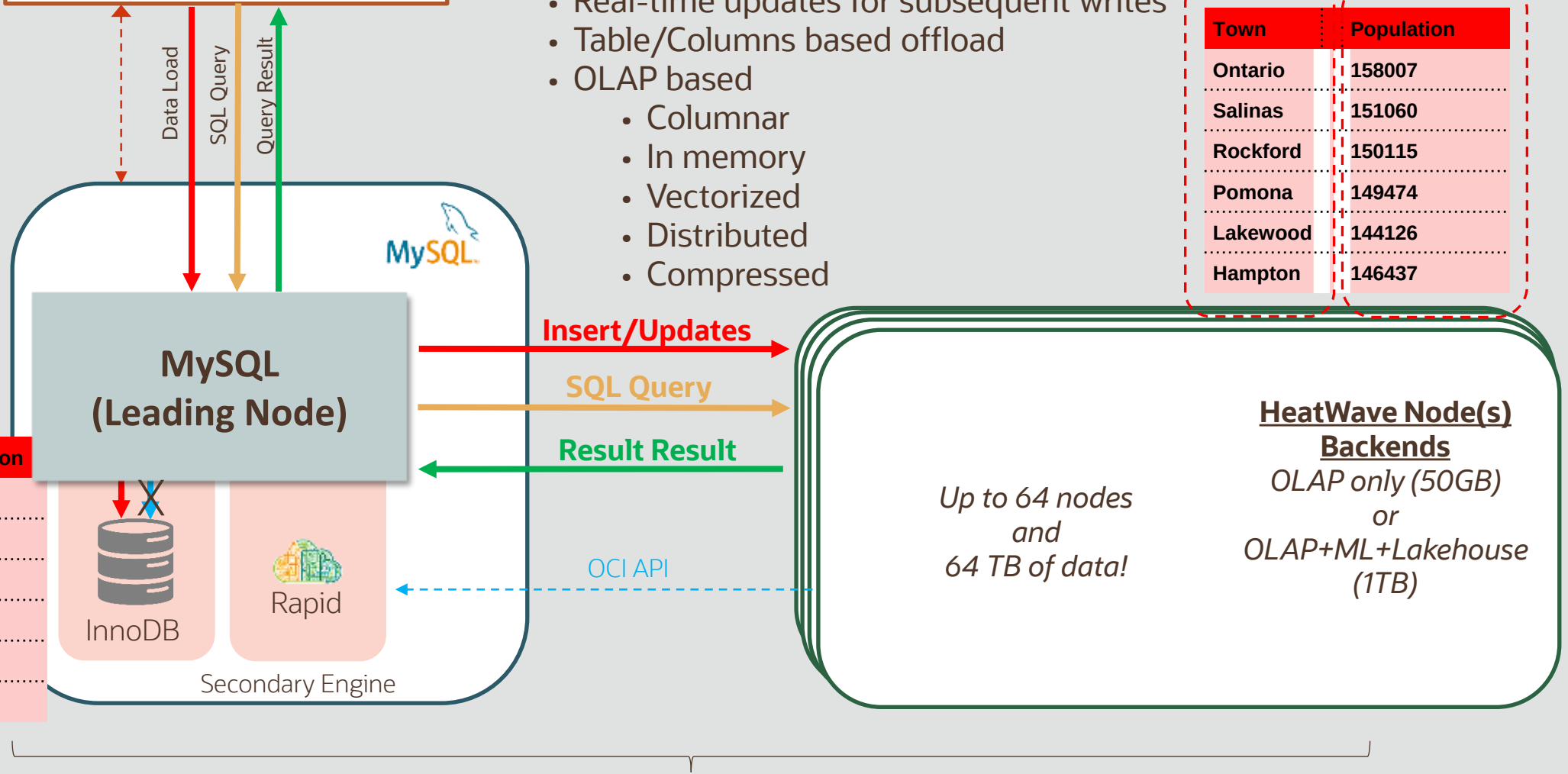
All MySQL App
MySQL Connector J, ODBC, NET,
Python, PHP, C, C++, Perl, Ruby

MySQL HeatWave

- Initial LOAD direct from InnoDB to RAPID
- Real-time updates for subsequent writes
- Table/Columns based offload
- OLAP based
 - Columnar
 - In memory
 - Vectorized
 - Distributed
 - Compressed

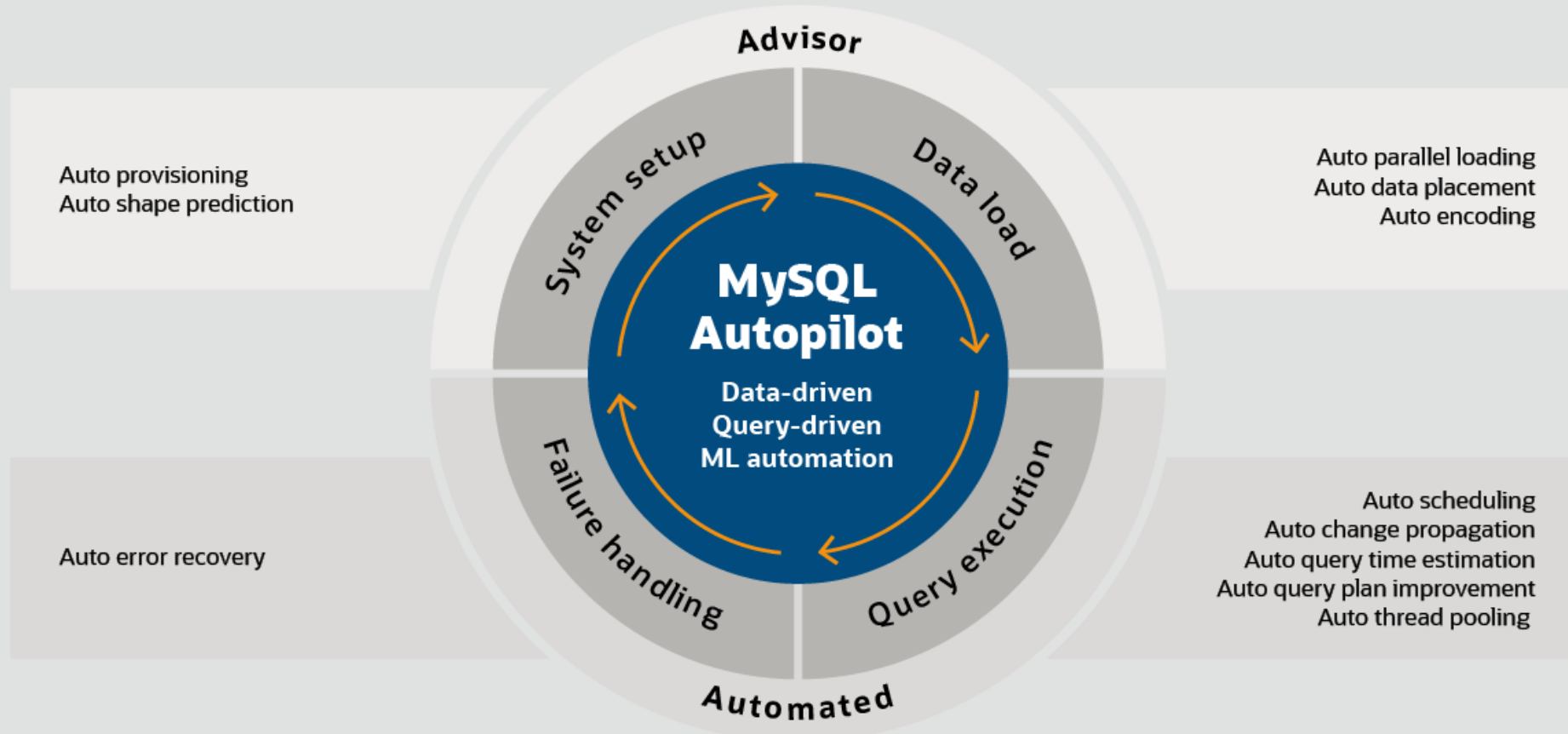
Town	Population
Ontario	158007
Salinas	151060
Rockford	150115
Pomona	149474
Lakewood	144126
Hampton	146437

Town	State	Population
Ontario	CA	158007
Salinas	CA	151060
Rockford	IL	150115
Pomona	CA	149474
Lakewood	CO	144126
Hampton	VA	146437



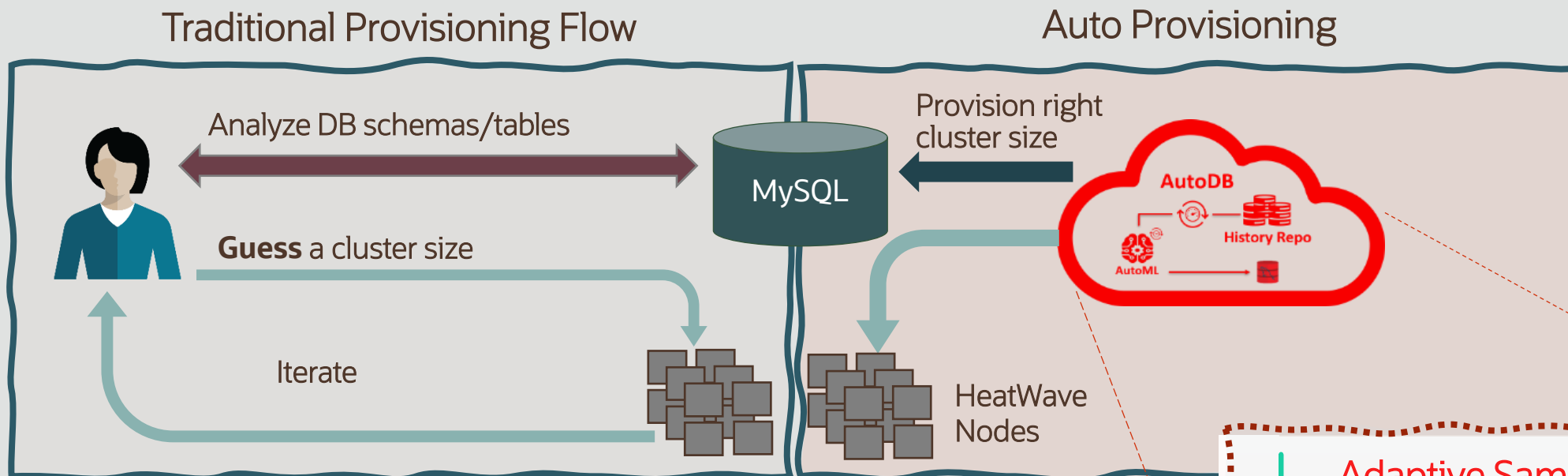
Machine learning-powered automation for MySQL HeatWave

High query performance at scale, higher OLTP throughput, and the best price performance

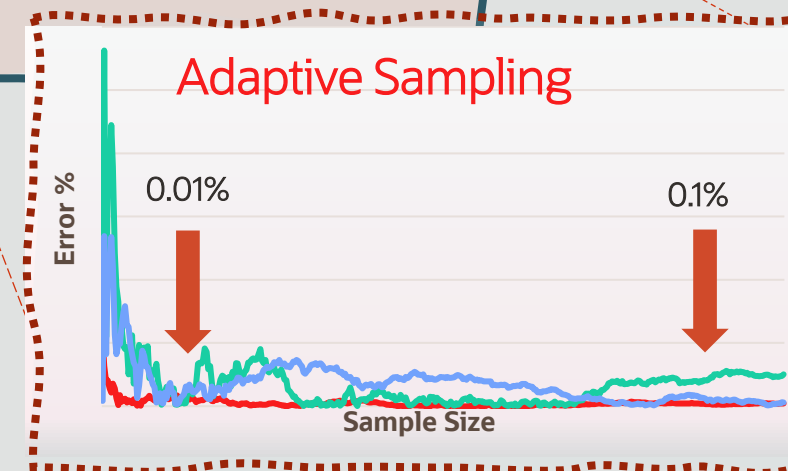


Auto Provisioning

Machine learning prediction of memory usage to estimate cluster size



Datasets	Accuracy in memory prediction
TPCH 1024G	98.4%
TPCDS 1024G	96.9%
Cust A	98.3%
Cust B	96.9%



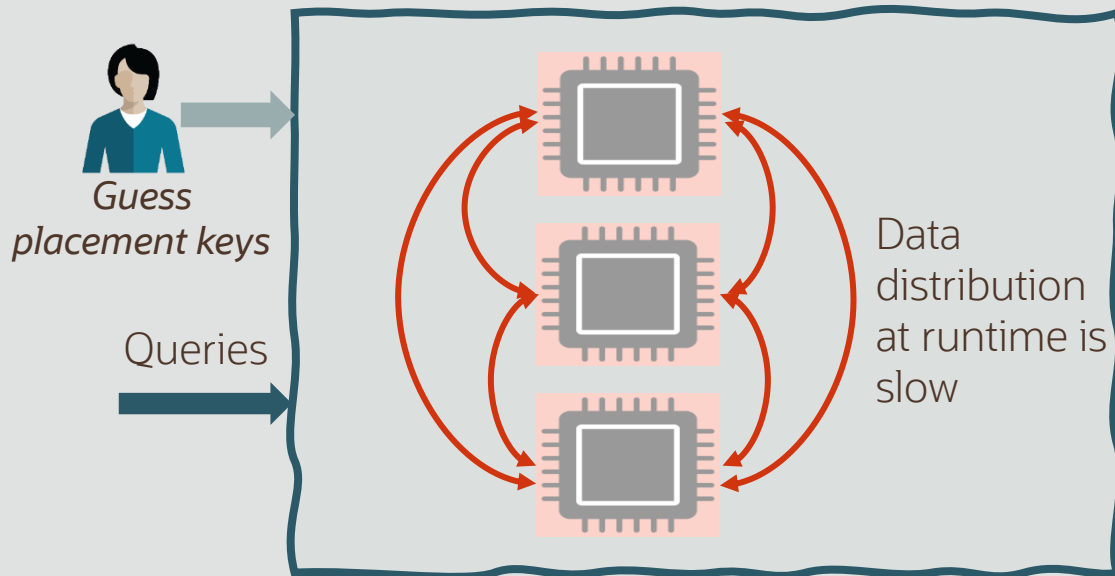
< 0.1% of data scanned for prediction

Auto Data Placement

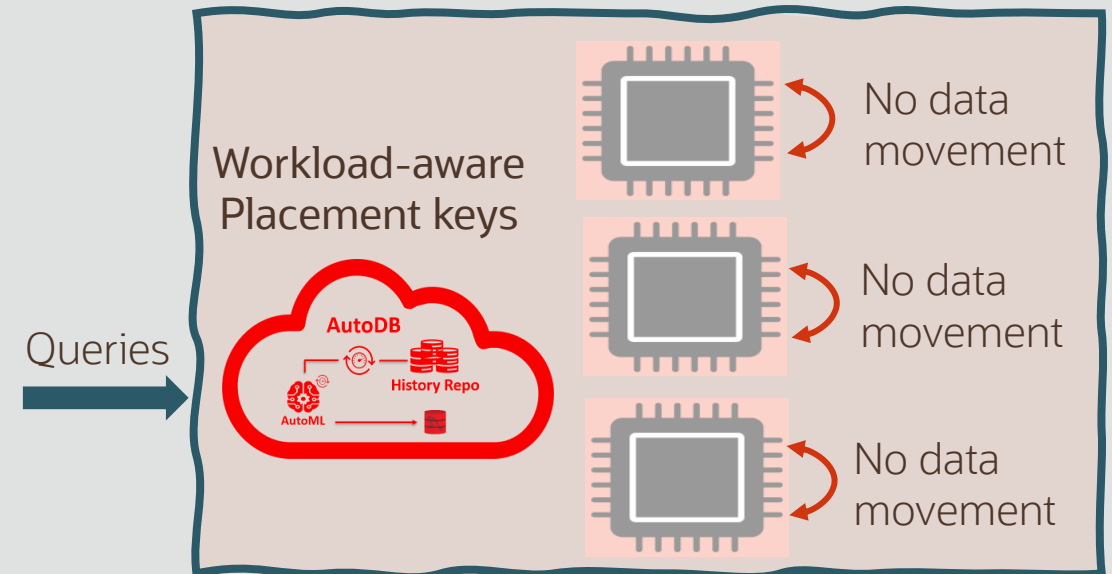
Machine learning prediction of optimal in-memory partitioning column

- System predicts optimal columns to partition data in-memory based on recent queries
- Predicts improvement in runtime

Manual Data Placement



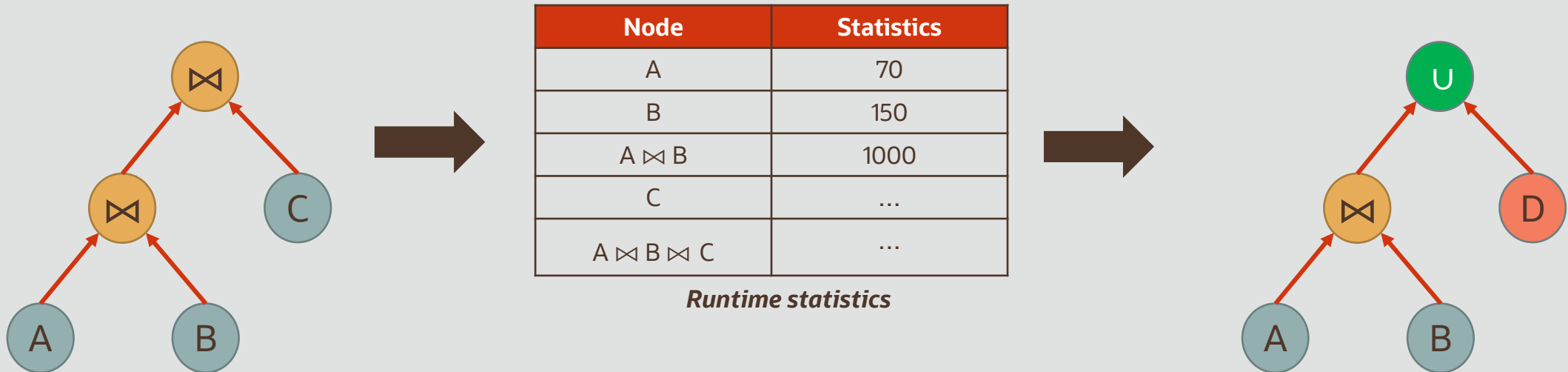
Auto Data Placement



Dataset	Time with Primary Key placement	Predicted Improvement with Autopilot	Actual Improvement from Autopilot
TPCH 1024	332 sec	26%	37%
TPCH 4096	373 sec	20%	25%

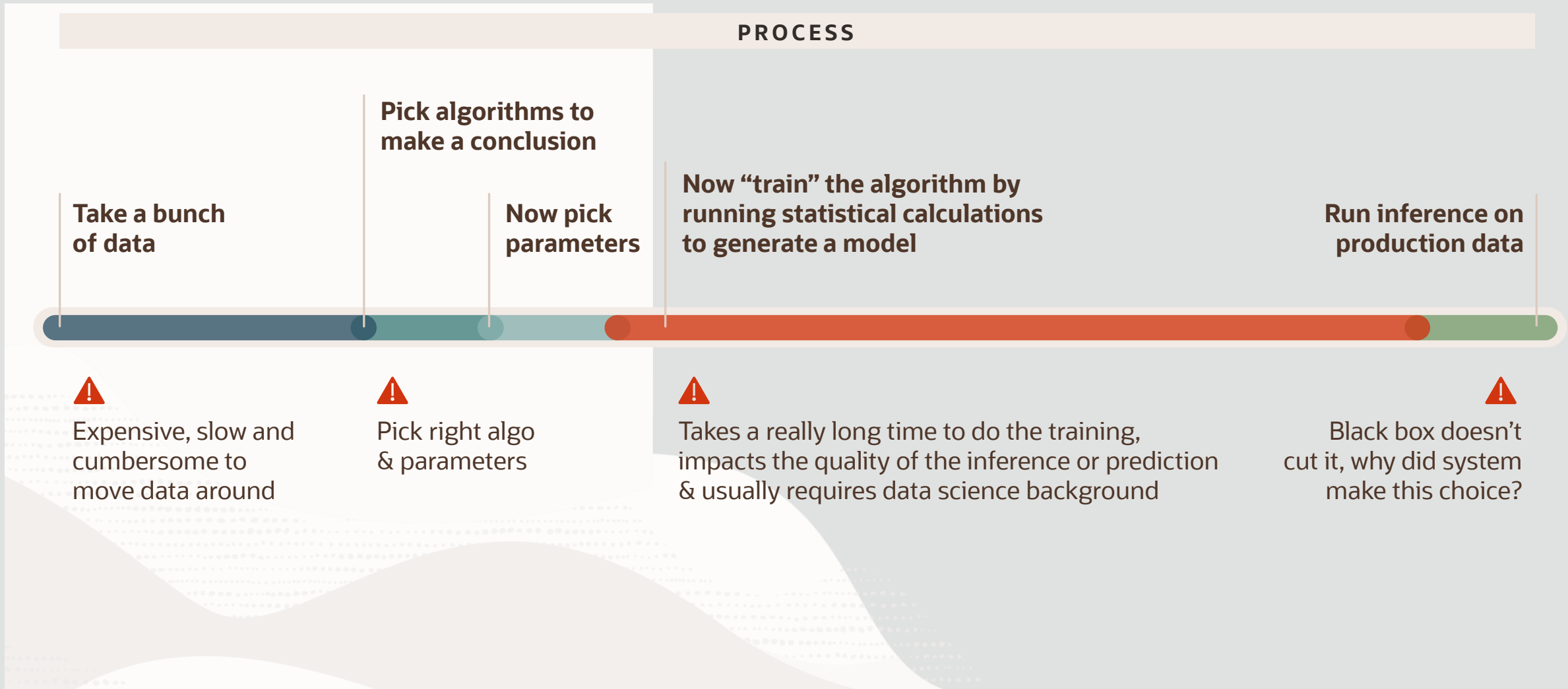
Auto Query Plan Improvement

Optimizer learns and improves query plan based on queries executed earlier



- Traditional caching techniques are not intelligent
- With Autopilot, system gets better as more queries are run
- For example, Autopilot improves TPCH, TPCDS 24TB performance by 40%

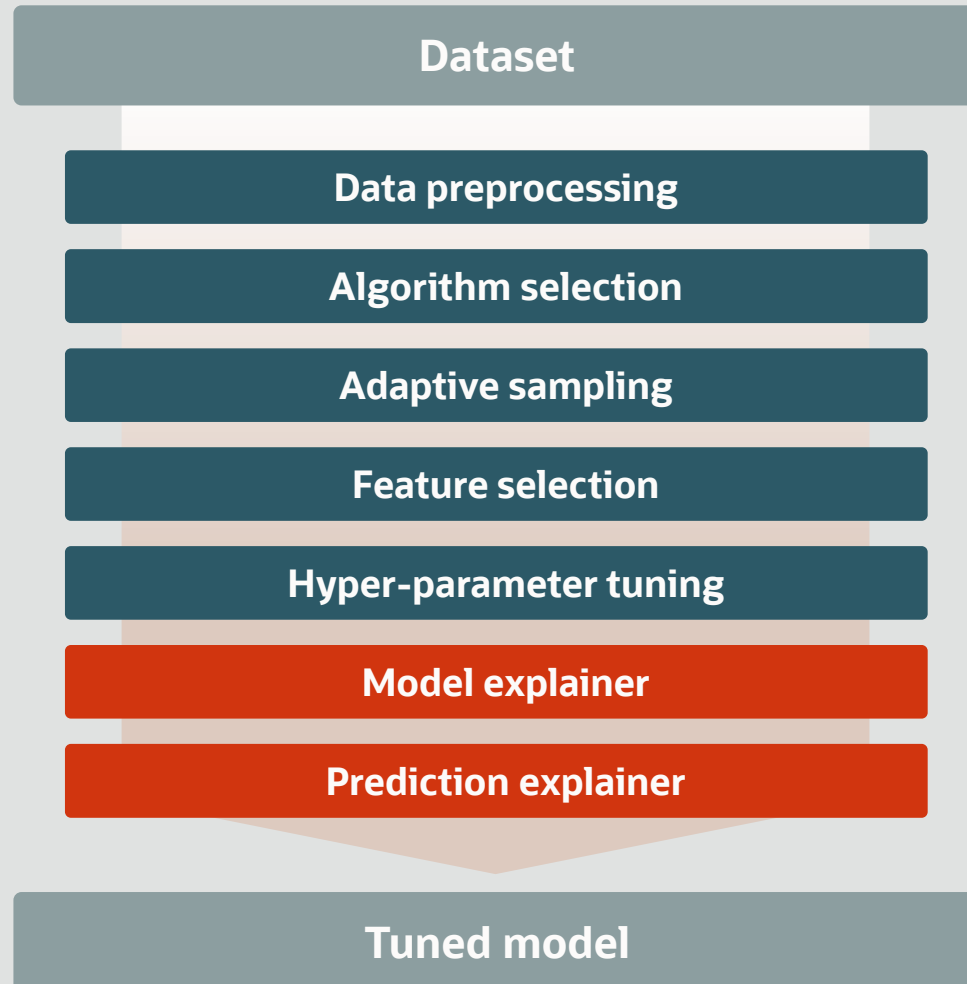
Machine Learning 101



Machine Learning 101



HeatWave ML



How is training in HeatWave ML fast?

- Training in HeatWave ML can be done in **one pass**
- **Highly parallel** hyperparameter tuning technique
- **Intelligent sampling** which doesn't compromise accuracy
- Training **automatically converges**
- **Scales** with the size of the cluster

HeatWave ML Explainability helps with

- Regulatory compliance
- Fairness
- Repeatability
- Causality
- Trust

Machine Learning techniques

- **Classification**

- Predictive modeling problem where a discrete value is predicted for a given input data
- **Examples:** Classify emails as spam or non spam, Customer churn prediction



- **Regression**

- Predictive modeling problem where a continuous value is predicted for a given input data
- **Examples:** Home prices prediction, Customer wait time in a queue prediction



- **Forecasting**

- Predictive modeling problem where a continuous value is predicted for time-ordered observations
- **Examples:** Demand Forecasting Based on Historical Sales Data, Commodity price forecasting (purchasing aspect)



- **Anomaly Detection**

- Predictive modeling problem where an anomalous value is predicted for a given input data
- **Examples:** Identifying anomalous packets in network security, Identifying anomalous sensor readings in an assembly line



- **Recommender System**

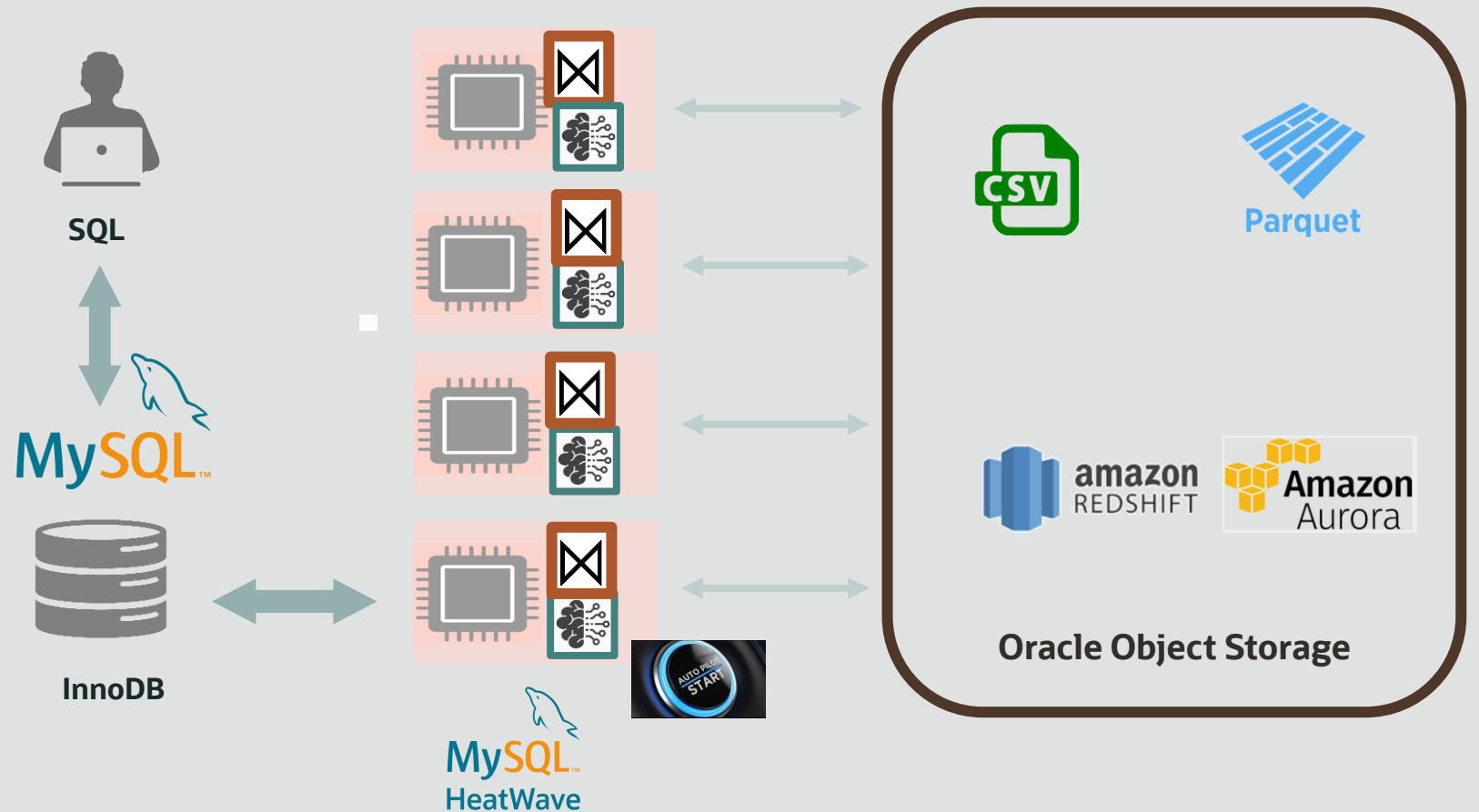
- Predictive modeling problem where a continuous value is predicted for a pair of user and item
- **Examples:** Movie recommendation, Click rate predictions



MySQL HeatWave Lakehouse

Support data on object store, multiple file formats, larger data size

- Data in InnoDB and HeatWave servers
 - OLTP
 - Analytics
 - Machine Learning
- + Data from Object Storage



Transaction Processing

Analytics

Machine Learning

Object Store

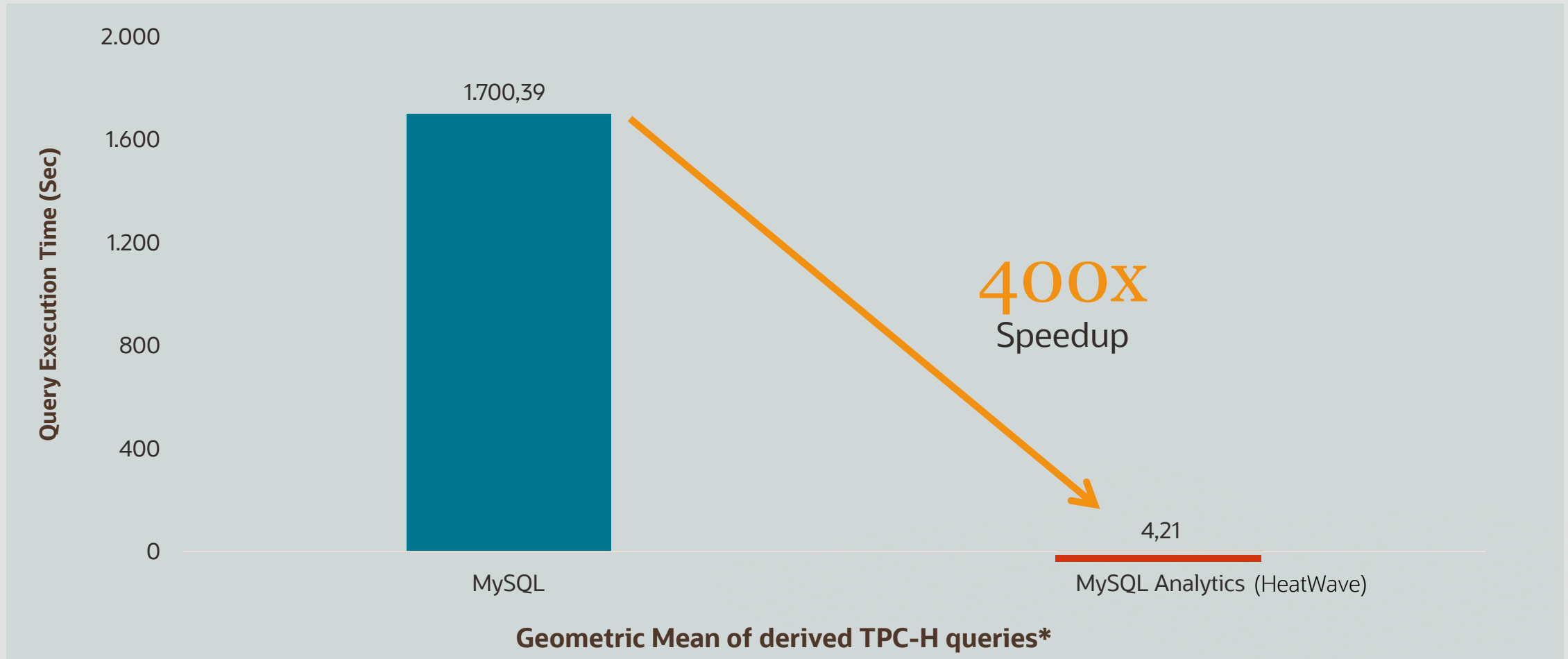
Lakehouse

MySQL HeatWave Shapes

- **Minimum configuration for OLTP**
 - 1 MySQL Frontend (16 OCPU, 512 GB RAM, 50 GB Storage)
- **Minimum configuration for OLTP+Analytics**
 - 1 MySQL Frontend (16 OCPU, 512 GB RAM, 50 GB Storage)
 - 1 HeatWave node (1 OCPU, 32 GB RAM, 50 GB Storage - can process up to 50 GB of data)
- **Minimum configuration for OLTP+Machine Learning+Lakehouse**
 - 1 MySQL Frontend (16 OCPU, 512 GB RAM, 50 GB Storage)
 - 1 HeatWave node (16 OCPU, 512 GB RAM, 50 GB Storage - can process up to 1 TB of data)
- **Biggest Shape for MySQL frontend**
 - 128 OCPU, 2 TB RAM
- **Up to 64 HeatWave nodes in OCI/Azure and 128 in AWS**

HeatWave dramatically speeds up analytic queries

400G, 64 cores



*Benchmark queries are derived from the TPC-H benchmark, but results are not comparable to published TPC-H benchmark results since they do not comply with the TPC-H specification

HeatWave Performance and Price Summary

TPCH

OLAP Performance

3x faster than Synapse
6.5x faster than Redshift
7x faster than Snowflake
10x faster than Big Query
1100x faster than Aurora
5400x faster than RDS

OLAP Price Performance

15x better than Synapse
13x better than Redshift
35x better than Snowflake
37x better than Big Query
2200x better than Aurora
7000x better than RDS MySQL

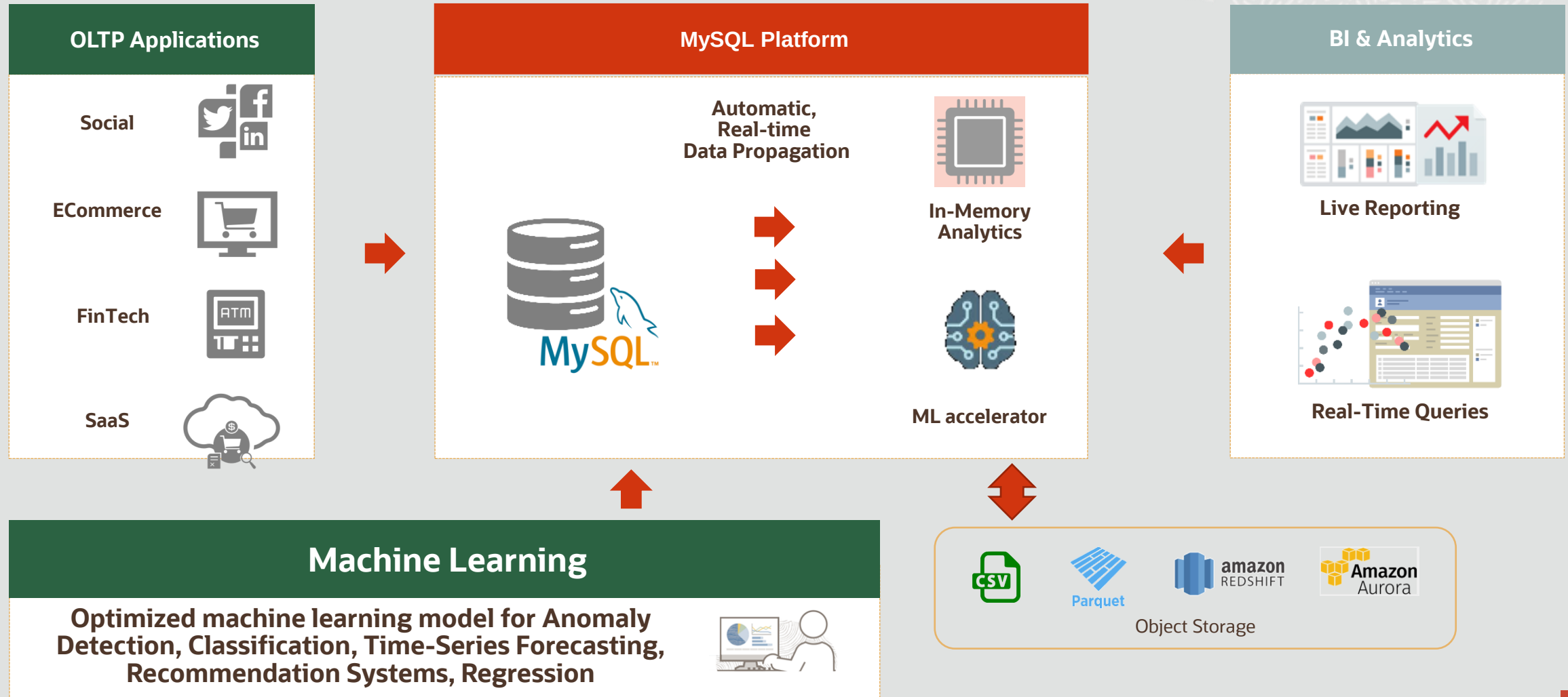
OLAP + OLTP

20x-500x faster than Aurora

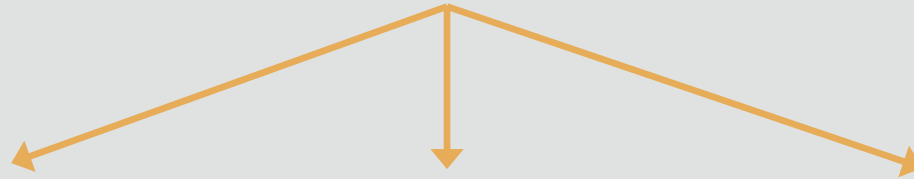
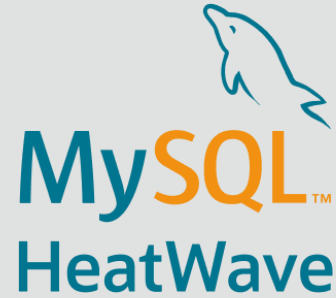
50x-1200x better price/perf
than Aurora

HeatWave

Single System for OLTP, Analytics and ML



HeatWave is Available in public clouds



ORACLE CLOUD
Infrastructure



NEW



NEW

Optimized to deliver the best price performance in each cloud

References

- MySQL HeatWave Database web page
 - <https://www.oracle.com/mysql/>
- HeatWave web page
 - <https://www.oracle.com/mysql/heatwave/>
- Oracle Cloud Cost Estimator
 - <https://www.oracle.com/cloud/costestimator.html>
- MySQL HeatWave Database Service documentation
 - <https://docs.oracle.com/en-us/iaas/mysql-database/index.html>
- Heatwave documentation
 - <https://docs.oracle.com/en-us/iaas/mysql-database/doc/mysql-analytics-engine.html>
 - <https://dev.mysql.com/doc/heatwave/en/>
- HeatWave performances comparison
 - <https://www.oracle.com/it/mysql/heatwave/performance/>



DEMO



9b. DEMO: MySQL HeatWave Cluster

Course Summary

What we have seen

- **We discussed the difference between MySQL Server Open Source and Commercial and the benefits of the partnership**
 - Introduction
 - Oracle MySQL Enterprise
- **We discussed and used OCI services**
 - Oracle Cloud Infrastructure overview
 - MySQL Database Service and HeatWave
- **We discussed and installed MySQL Server, community and Enterprise features**
 - Installation and Architecture
 - MySQL Security
 - Database Design
 - MySQL Enterprise Monitor
 - MySQL Backup
 - Replication and High Availability
- **MySQL HeatWave, the MySQL DBaaS in OCI**

Next steps

- **Complete the evaluation form of the workshop**
 - You can also use OCI CLI to easily automatize stop & start using crontab
- **Study for the exam 1Z0-921 - MySQL 2021 Implementation Essentials**
 - Workshop slides
 - MySQL Documentation online
 - If you fully attended this bootcamp, call your Channel Rep to check if you are eligible for a free voucher
- **Don't forget to practice**
 - You can ask an OCI TRIAL active for 30 days/300\$ at <https://www.oracle.com/cloud/free/>
 - Don't forget to power off compute instances and/or MySQL HeatWave instances when you don't use them
 - You can download the MySQL Enterprise software in trial (30 days) from <https://edelivery.oracle.com>
- **For your daily work with MySQL Commercial, on-premise and cloud, the MySQL team can help you**
 - Channel, Sales teams, Sales Engineers, Support, etc.

Enjoy your journey with  **MySQL™** !



Thank you!





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